Application Guidebook

Application and installation guidelines
for high-speed MTU Onsite Energy gas engine-generator sets

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1 General Information

1.1 About this manual

This application engineering manual describes the specification and application of systems for generating electricity, heat and/or cold from MTU Onsite Energy. The main purpose of this manual is to support standard system business for high-speed gensets.

Structure of the manual

The project planning manual corresponds to the sequence of a project - starting with the request for quotation, through consultation and on to project processing. The project planning manual consists of the general section "Basic knowledge - general" and the specific section "Application engineering". The first section explains basic facts that are important when planning power and heat generation systems. The second section is divided into the application engineering of gas gensets and application engineering of diesel gensets. Both have the same structure in the following subsections:

- Basic knowledge
- Customer requirements (request for quotation)
- Planning criteria
- Engine-generator set - components
- Infrastructure
- Transportation
- Validation and commissioning

In the subsection Basic knowledge, the basic terms and components of a system for electricity, heat and/or cooling energy generation are explained. The various operating modes are presented, and the general conditions as well as operational limits and legal specifications are stipulated. A general explanation is provided as to how a genset or a plant from MTU Onsite Energy is operated and which energy forms are made available. A reference is already made here to aspects that are important for subsequent planning.

The subsection Customer requirements explains in detail the typical customer requirements of a system for electricity, heat and/or cooling energy generation. Factors that influence the performance of the system are presented.

In the case of a project, the recorded customer requirements are adopted in the planning of the genset or plant. The selection options of the genset components and the infrastructure are listed in the following subsections.

The function of the components available for selection is explained in the subsection Engine-generator set - components. Here, the interconnections that have to be observed when planning complex plants are presented.

In the Infrastructure subsection, the assemblies and interfaces around the genset are explained. Each genset, whether a stand alone or part of a plant, requires a certain infrastructure to function correctly.

Products from MTU Onsite Energy are configured as fixed installations or mobile gensets. For this reason, a separate chapter has been dedicated to transport with information on special features and safety aspects.

The project is completed with validation and commissioning of the machine. This chapter not only lists the basis sequences and steps that have to be observed, but also refers to documents and checklists offered by MTU Onsite Energy to guarantee a smooth process.

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Target group

The project planning manual is intended for system integrators, MTU distributors, MTU service partners and MTU project planners. It applies to all worldwide available high-speed MTU Onsite Energy systems for electricity, heat and cooling power generation.

An assumption is made that readers have basic knowledge of technology and physics.